BEM 103

Class 2:
Modern Finance

Part 1 Financial literacy

- Returns uncertainty
- Stocks vs bonds

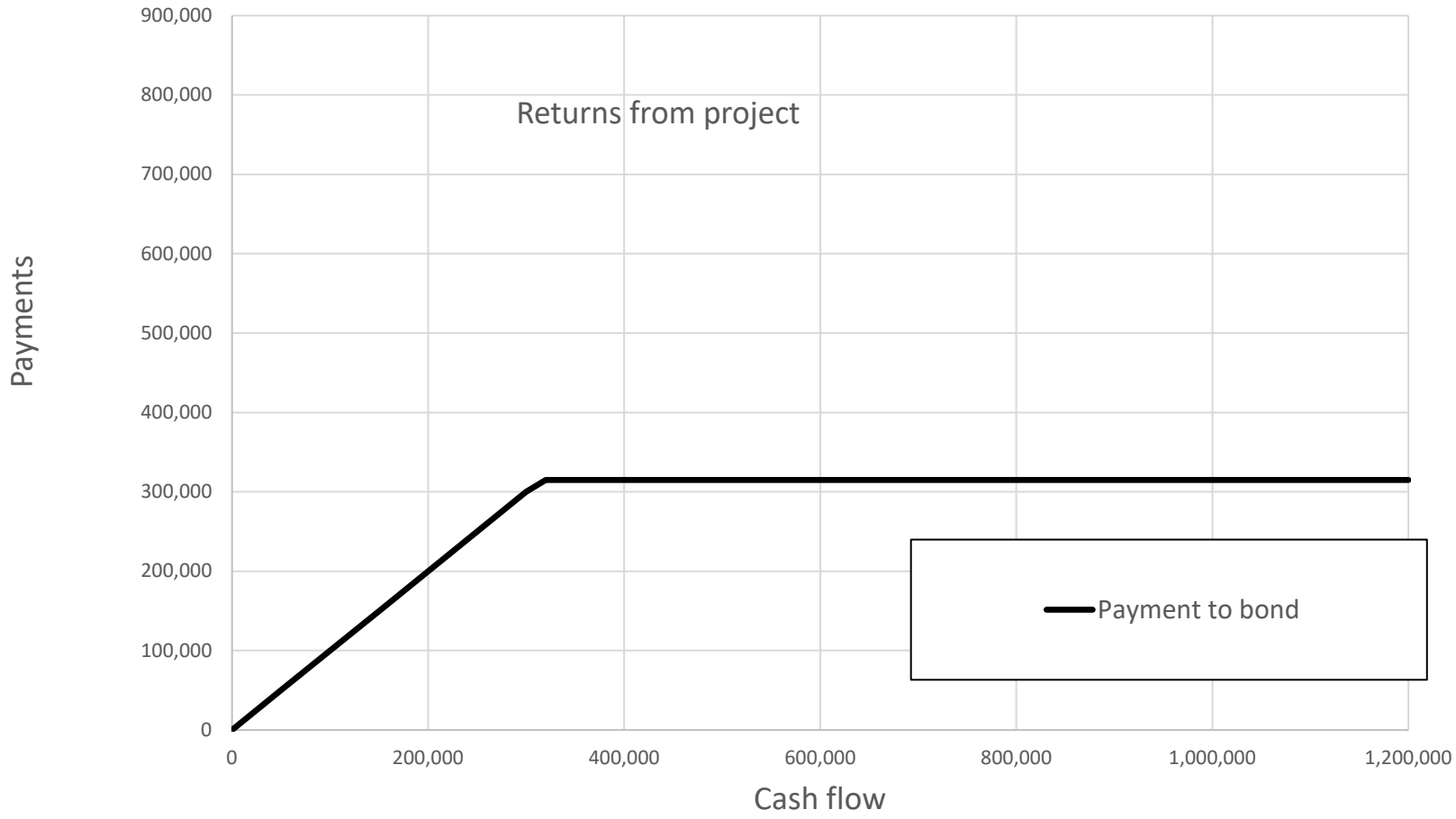
Part 2 Financial axioms

- Financial markets are competitive:
- Value additivity;
- No Free Lunch;
- Markets are efficient

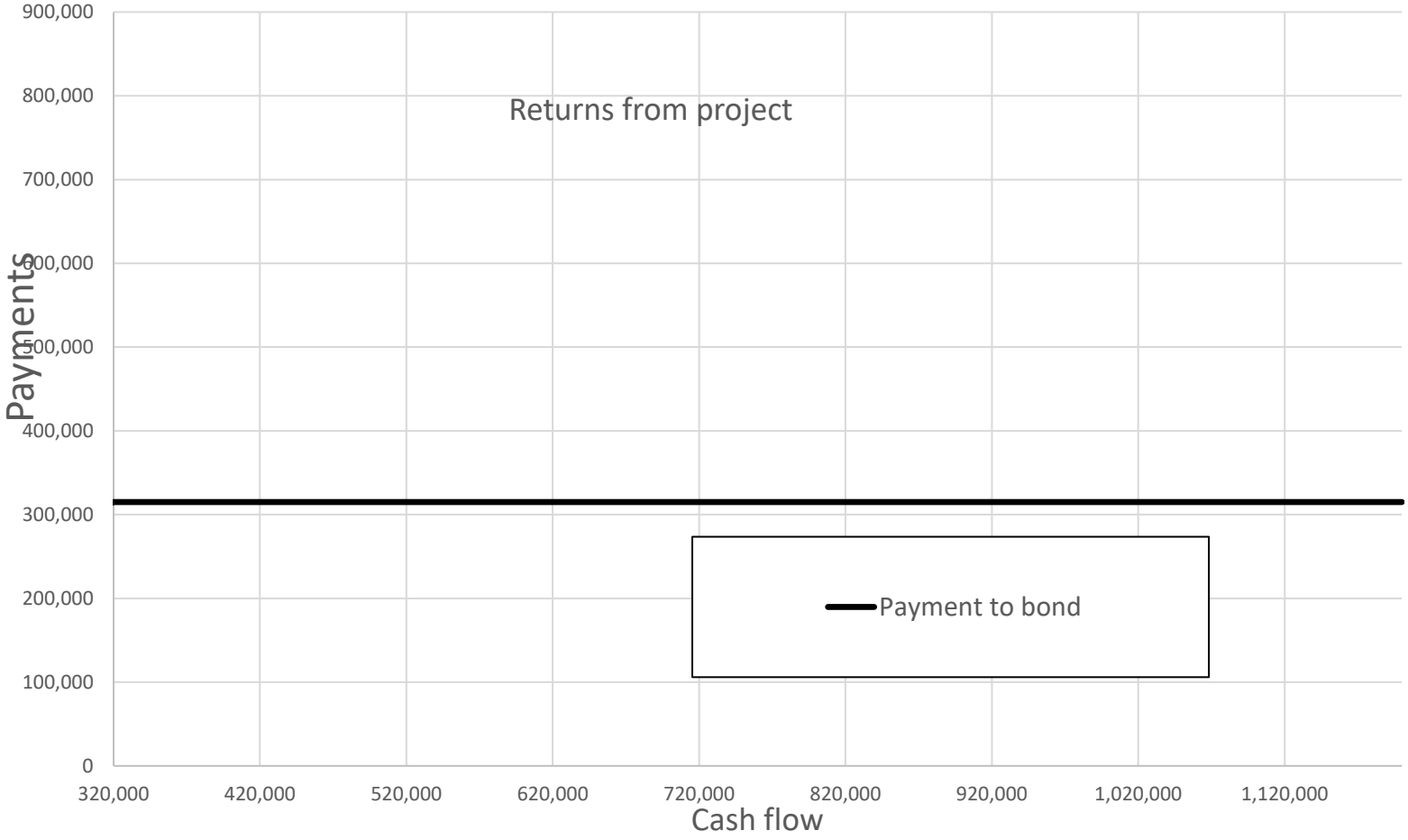
Part 3 behavioral finance and imperfect markets;

- Market impact
- Behavioral finance

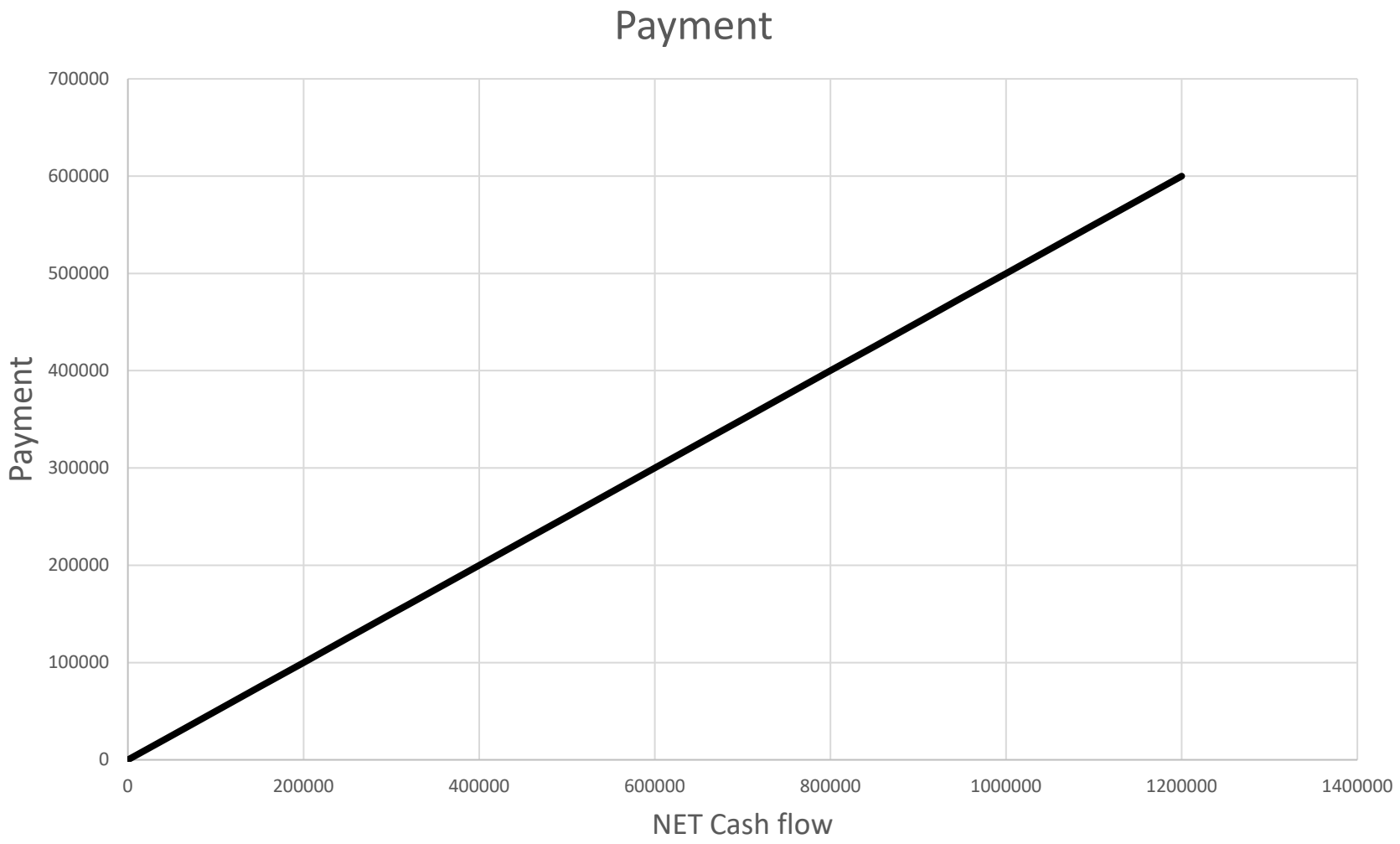
Debt and equity
part 1 a bond



Debt and equity
part 1b a riskless bond

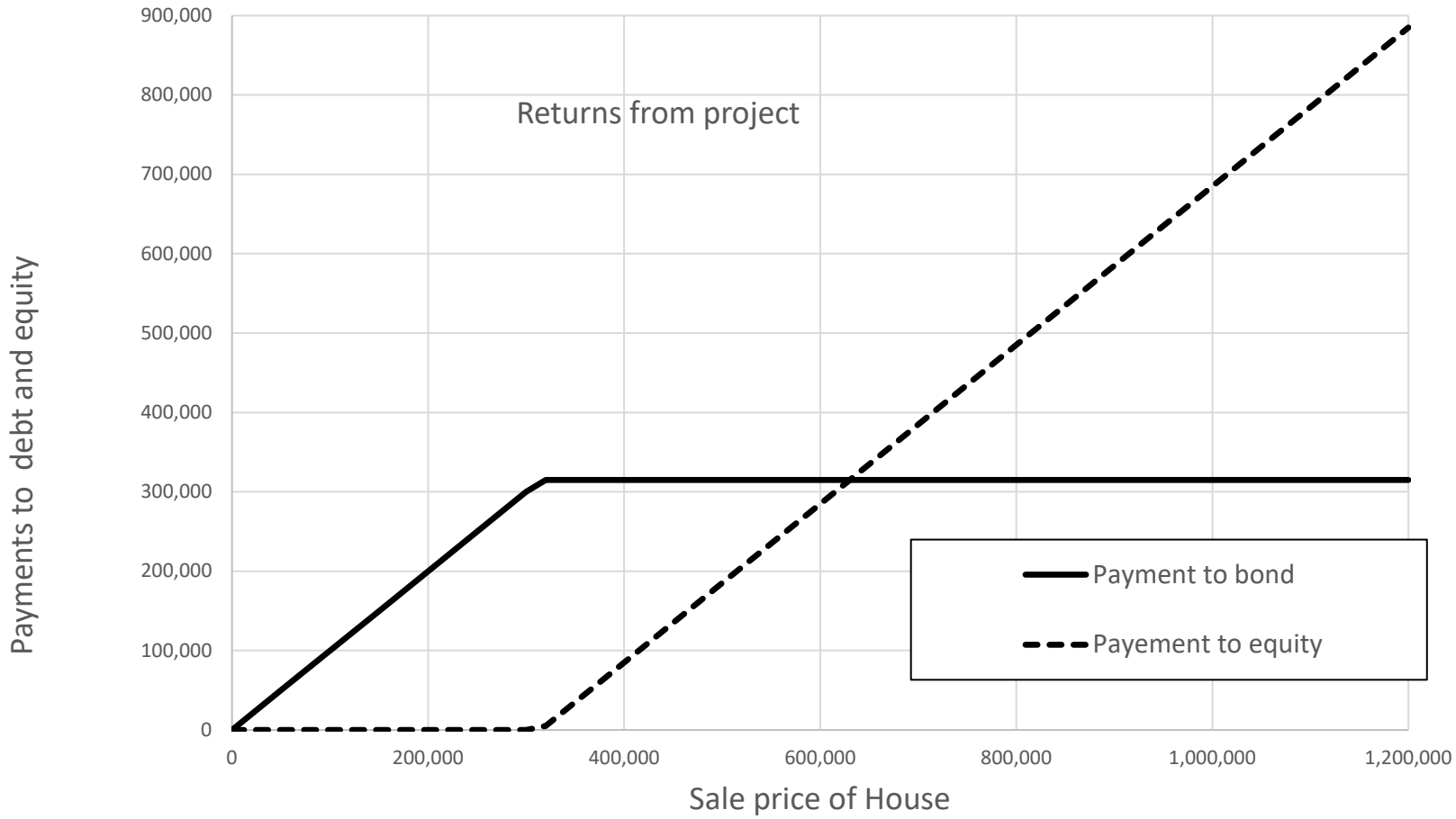


Debt and equity
part 2 Equity



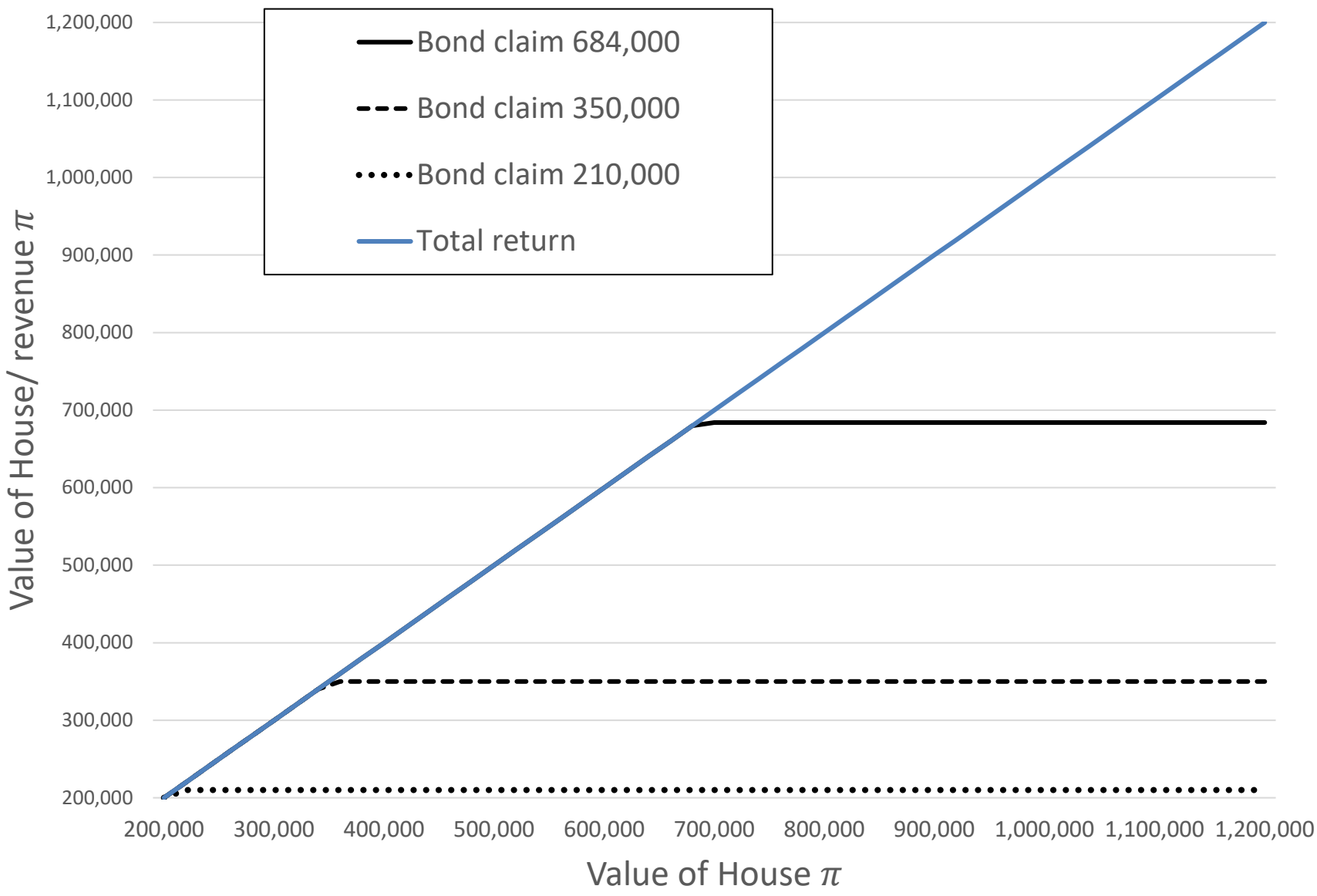
Debt and Equity

- Debt is a promise to pay B in return for getting L today
  - For example L and B=L(1+r)
  - Get paid depends on the project cash flow π
  - B if $\pi > B$ π if $\pi < B$
- Equity is a share of the net cash flow
  - Pay L to secure a share s≤1 of the net cash $(\pi - B)$
  - It varies!
  - s$(\pi - B)$ is the investor return
- Participation constraint
  - Is investor willing to invest L to b share S ?
  - Is investor willing to pay L to buy claim B ?Debt and equity matter
  - Not to the cash flows but because of distribution of returns


Returns from project
Payments to debt and equity
900,000
800,000
700,000
600,000
500,000
400,000
300,000
200,000
100,000
0
0
200,000
400,000
600,000
800,000
1,000,000
1,200,000
Sale price of House
Payment to bond
Payement to equity

Debt and Equity

- If there is no risk then debt and equity are the same. π is known.
  - Lend at most L=$\pi/(1+r)$ and in all cases return is r.
  - Investor can invests until her share is 100% or V=$\pi/(1+r)$
- If there is risk then debt and equity are different
- Different levels of debt affect returns and who gets them *once* the return is realized.


Bond claim 684,000
Bond claim 350,000
Bond claim 210,000
Total return
1,200,000
1,100,000
1,000,000
900,000
800,000
700,000
600,000
500,000
400,000
300,000
200,000
Value of House/ revenue π
200,000
300,000
400,000
500,000
600,000
700,000
800,000
900,000
1,000,000
1,100,000
1,200,000
Value of House π

Two sides of Modern finance

- Two core and opposing views
- Efficient markets hypothesis
  - All the current information about future returns is contained in the price
  - You can’t beat the market except by luck
  - That you beat the market today does not increase your chances tomorrow
- Behavioral finance/imperfect markets
  - Individuals have systematic biases in how they handle financial information and decisions
  - Intermediaries have a personal profit motive that leads them to systematically distort the market
  - Both problems are worst in booms and in busts
- Efficient market hypothesis, no one can beat the market,
- Imperfect markets, only insiders can beat the market

Axioms of modern finance

- **A1 Financial markets are competitive**
- **A2 Value additivity;**
- **A3 No Free Lunch;**
- **Implication: Markets are efficient**
  - **Equilibrium where capital is allocated to most efficient use.**
- **Need definitions and look at these from three perspectives**
  - **Economics**
  - **Finance**
    - **Secondary Markets**
    - **Primary Markets**

A1. Financial markets are competitive

- What is a competitive market?
- Economics
  - Individual actor decisions do not matter to price (or quantity)
  - Price = marginal cost Profits are zero
  - Supply curves are flat
- Finance
  - Secondary markets
    - Marginal cost is not a relevant concept
    - But if markets are competitive
      - Actors take prices as given (they do not affect price).
      - Size of orders does not matter.
      - No arbitrage ⇔ profits are zero
  - Primary markets
    - Investment bank fees reflect costs

A1. Financial markets are competitive

- If A1 is true then it makes many decisions easy
  - Actors do not need (want) to be strategic. Price is the only statistic that matters
  - The only issue for investment then is how much risk do you want to bear
- If A1 is false, you have to be careful in the market.
  - Your strategy matters to the strategies of other players
  - Other participants may have advantages you do not have.

A2 Value additivity

- Economics
  - Given competition cost of a bundle of goods cannot be less than the cost of the most expensive unit and it cannot be more than the sum of the unit costs.
  - Bounds (because there may be some economies of scale or scope)
- Finance A2 is stronger.
  - Price of a basket of securities is equal to the sum of the prices of individual contracts times the quantities ⇔ no arbitrage condition.

A2 Value additivity

- Suppose there are n securities each with price p_i
- Let X be a portfolio $X=\{x_1, x_i,\ldots, x_n\}$
  - x_i is the number of shares or bonds in the portfolio of type i.
  - V_X is the value of the portfolio
- Suppose $V_X > p_1x_1+\ldots+p_1x_i+\ldots+x_np$.
  - You can arbitrage. Sell the portfolio, buy the individual securities and you will make money.
- Suppose $V_X < p_1x_1+\ldots+p_1x_i+\ldots+x_np$.
  - You can arbitrage. Buy the portfolio and break it up and you will make money.

A3 No Free Lunch

- Economics
  - Can't sell nothing for something
- Finance
  - Asset that has a zero payoff for certain has zero price. (think of the price of a bond that matures in 2022, what should you be willing to pay for it in 2026?)
  - Another no-arbitrage condition.
  - Suppose x_i has a zero payoff. Compare $X=\{x_1,\ldots,x_i,\ldots.,x_n\}$ and $O=\{x_1,\ldots,0,\ldots,x_n\}$
  - $V_X > V_O$ violates the no arbitrage condition
  - Sell V_X buy V_O

If A1, A2, A3 hold Financial Markets are Efficient

- Economics
  - A market allocation is Pareto efficient
  - You cannot find an alternative allocation that makes someone better off without making someone worse off (gains of trade have been exhausted.
    - Allocation distribution of what ever goods or services are under consideration

Financial Markets are Efficient

- Finance
  - Prices in financial markets correctly aggregate all the available information. They are the best forecast of future income flows.
    - Does not mean they correctly predict every point in the future.
    - prices reflect the true distribution of future cash flow (these could be high or low but the market knows how likely they are to be high or low).
- Systematic biases offer arbitrage opportunities to those who detect then and thus they will be corrected
  - Financial markets are efficient in equilibrium

Implications

- If “on average at least” the axioms of finance hold

AND

- The statistical past is the statistical future

Then price histories are sufficient statistics for making long-term investment decisions

- There are no dollar bills on the ground
- You do not need to waste your time on research, or pay an advisor because competition has already squeezed all the profits out of the market
- You may want to use an advisor or do research because you have specific time or risk preferences

Some evidence in favor of Efficiency Fama and French J of Fin 2010

- "In short, suppose that when returns are measured before costs (fees and other expenses), passive investors get passive [market average] returns. This means active investment must also be a zero sum game…*before* costs. If some active investors have greater than market returns *before costs, it is dollar for dollar at the expense of other* active investors. After costs, that is, in terms of net returns to investors, active investment must be a negative sum game"
- Because active investors make idiosyncratic portfolio choice some, by chance or luck, must have above market returns.

- Fama and French "use long histories of individual fund returns and bootstrap simulations of return histories to infer the existence of superior and inferior funds. We compare the actual cross-section of fund returns relative to market, *to the results from 10,000 bootstrap simulations of the cross-section. The returns of the funds* in a simulation run have the properties of actual fund returns, except we set true *α to zero in the return* population from which simulation samples are drawn.
- The simulations thus describe the distribution of *α* estimates when there is no abnormal performance in fund returns. Comparing the distribution of *α estimates* from the simulations to the cross-section of *α estimates for actual fund returns allows us to draw inferences* about the existence of skilled managers.
- For fund investors the simulation results are disheartening. When *α is estimated on net returns to* investors, the cross-section of precision-adjusted *α estimates, t(α), suggests that few active funds produce* benchmark adjusted expected returns that cover their costs.
- Thus, if many managers have sufficient skill to cover costs, they are hidden by the mass of managers with insufficient skill. On a practical level, our results on the long-term performance say that true *α in net returns to investors is negative for most if not all active funds,* including funds with strongly positive *α estimates for their entire histories."*

- Note this is a necessary condition for market efficiency but not sufficient? Why?

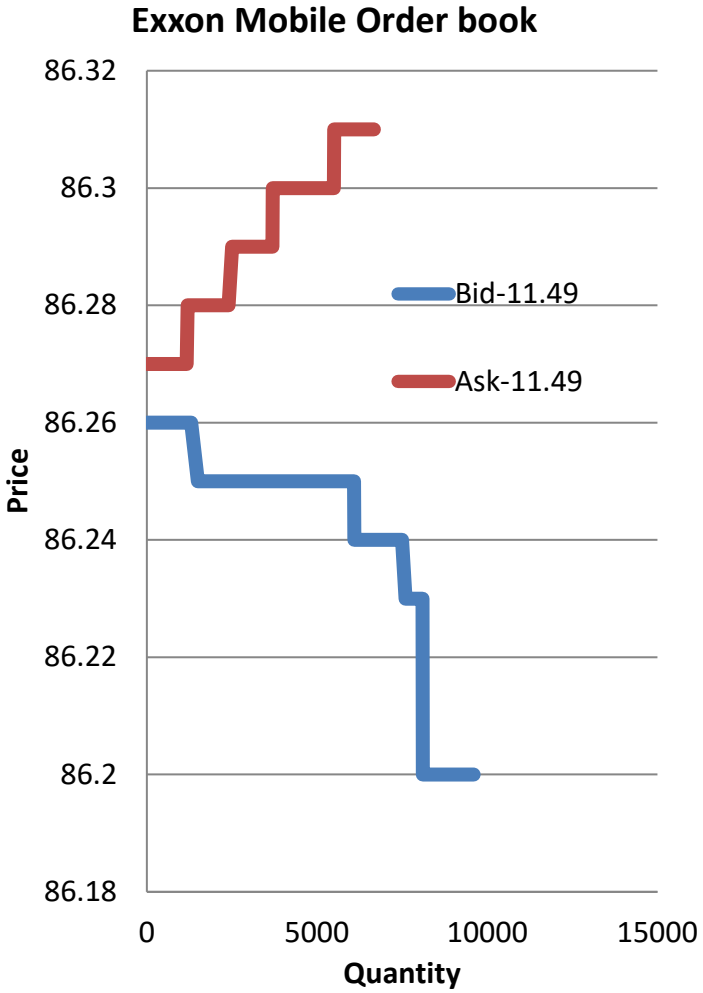
Some evidence against Efficiency

- Hedge funds
- High frequency trading
- Warren Buffet
- Other arbitrage intermediaries

- Problem is
  - Tending towards efficiency is not the same as efficiency

A1. Financial markets are competitive but order books are not flat

- Is this a violation of competition?
  - Supply curve is definitely not flat
- In fact this is a major problem for large players because their orders move the market (A lot!)



Lack of competition

- Markets are thin
  - Sell 1000 shares of Apple in 5 minutes at the current price $175 probably easy (15 billion shares outstanding, 42.6 million shares trade a day) add to demand but no much (0.2%)
  - Sell 1000 shares of Dish in 5 minutes at the current price $5.6 probably harder (500 million shares outstanding, 5 million shares trade a day) add to demand (2% in an average 5 minutes of trading)
  - Sell 1000 shares of Consensus Cloud Solutions in 5 minutes at the current price $26.6 probably harder (19 million shares outstanding, 114,000 shares trade a day). You double demand.
  - Market impact. Most markets are closer to CCS than to Apple. This limits arbitrage.
  - You want to sell 1000 Shares of CSS (price $26) and buy 928 shares of Janux Therapeutics ($28) expect to get less for your shares than the price for CSS and pay more than the price for JANX.
- Some players have advantages
  - Top 20 banks in the US hold more than 90% of all deposits
  - Mutual fund industry is also very concentrated
    - Have access to data for strategies that others do not.
    - Note this does not imply insider trading
  - Too big to fail goes both ways
- Return to these issues later in the quarter.

A2 Value additivity
Bond Stripping

- 1 year T-Bill. Create two assets
  - Own B bills
  - A_1: the interest coupons
  - A_2: the final payout
  - No free lunch says $P_B B = P_1 A_1 + BP_2 A_2$
- If true, why do we create these assets?
  - We do so there must be a demand (and a return)
  - Initially the return was high, but then market forces drove profits down
  - And the federal government started issuing its own strips.
- Heterogeneity in demand?
- Is No Free Lunch an aggregate statement or a marginal one?
- Problem is between theory and the real world
  - in the real world some actors have a high willingness to pay for a particular asset and supply is relatively small so a premium exists
  - Violation of competition

https://www.treasurydirect.gov/research-center/timeline/strips/

Closed End Funds

- Fund is set up with a fixed number of shares and are precursors to ETF (exchange traded funds)
- NAV (net asset value) is not always equal to MP.
- NAV is the cost of replicating the portfolio
- MP is the price on the market

- **Nuveen Real Estate Income Fund (JRS)**
  - $220 million in assets. NAV $8.34 MP $7.63
- **Tortoise Energy Infrastructure Corporation (TYG)**
  - $875 million in assets. NAV $44.8 MP$41.8
- Notice that the underlying shares are not liquid so if there was to be a difference it should go the other way (the fund being more liquid than the stocks)
- Notice also that if you wanted to ‘correct’ the market you would have to buy up the shares (so you have to have deep pockets) and set up your own “fund”

Next time

- <u>Class 3: Time is money</u>
- **Discounted present value**
- **Rates of return**